

Manipal School of Information Sciences

AME 5151 - Applied Linear Algebra Lab

Assignment 1

Academic Year: 2026-2027, Semester: 1

Friday, 11 September, 2026

This assignment maps to the lab course outcome CO1:

Apply Python's legacy libraries for coding vector and matrix-related operations.

Total Marks: 5

Use your implementation of the `Vector` class. Implement the following methods and write unit tests for them in a separate file named `test_vector.py`.

1. (1 point)

`mean` method that returns the mean of the vector entries. The mean is computed as the sum of all entries divided by the number of entries.

2. (1 point)

`demean` method that returns a new vector representing the de-meanned version of the original vector (*self*). The de-meanned vector is obtained by subtracting the mean of the original vector from each of its entries.

3. (1 points)

`std` method that returns the standard deviation of the vector entries.

The standard deviation is computed as the square root of the average of the squared deviations from the mean. Use the `demean` method to help with this computation.

4. (2 point)

Write meaningful unit tests for your implementation of the above methods. The unit tests must cover properties of the mean, de-meanned vector, and standard deviation mentioned in the lectures. You can refer to the lecture notes/slides for guidance on what properties to test.